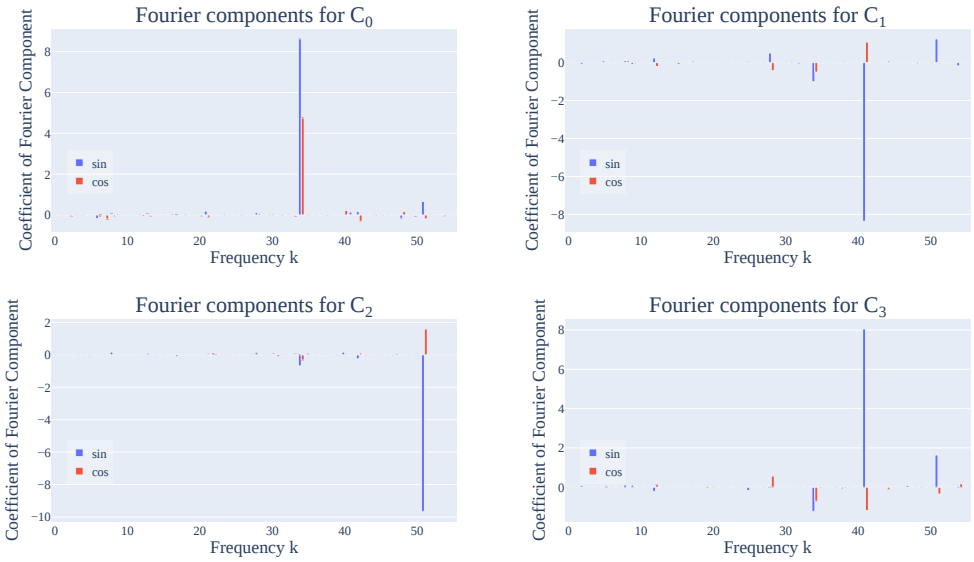

Figure 10: We plot the attention pattern weights C_j in the Fourier basis for each of the four heads $j \in \{0, 1, 2, 3\}$. We observe significant sparsity, with almost all of each term being associated with a single frequency.

Mechanistic Analysis of Attention Patterns. We can further mechanistically analyse *how* the model achieves this form. The following is a high-level sketch of what is going on:

First, note that the attention score on position 0 and head j is just a lookup table on the input token a (of size P). To see why, note that $A_0^j = m{x_0^{(0)}}^T {W_K^j}^T W_Q^j x_2^{(0)}$. $x_2^{(0)}$ is constant since the token is always '=' and $x_0^{(0)} = W_E t_0 + p_0$. So this reduces to $t_0 \cdot C_j + D$ for some constant vector $C_j = W_E^T {W_K^j}^T W_Q^j x_2^{(0)} \in \mathbb{R}^p$ and some scalar $D = p_0^T {W_K^j}^T W_Q^j x_2^{(0)}$. As t_0 is one-hot encoded, this is just a lookup table, which we may instead denote as $C_j[a]$

Next, note that the attention pattern from $=\to 0$ is $\sigma(C_j[a] - C_j[b])$. As argued in Appendix A.1, the attention paid $=\to=$ is negligible and can be ignored. So the softmax reduces to a softmax over two elements, which is a sigmoid on their difference. As form of C_j does not mention the token index or value, it is the same for position 0 and 1.

We now show that C_j is well-approximated by a wave of frequency w_{k_j} for some integer k_j. That is, $C_j[a] \approx F_j \cos(w_{k_j} a) + G_j \sin(w_{k_j} a)$. We do this by simply computing C_j and fitting the constants F_j and G_j to minimize ℓ_2 loss, and display the resulting coefficients for each head in Figure 10. This fit explain 99.02%, 95.21%, 99.10%, 92.42% of the variance of C_j respectively. Interestingly, the coefficients of heads 1 and 3 are almost exactly the opposite of each other.

For each head j, $\sigma(C_j[a] - C_j[b]) \approx 0.5 + E_j(C_j[a] - C_j[b])$ for some constant E_j—that is, the sigmoid has some linear approximation. (The intercept will be 0.5 by symmetry.) The striking thing is that, because the inputs to the sigmoid for the attention heads are over a fairly wide range ($[-5, 5]$ roughly), the linear approximation to the sigmoid is a fairly good fit, explaining 97.5% of the variance.

We validate that this is all that is going on, by replacing the sigmoid with the best linear fit. This *improves* performance, decreasing test loss from $2.41 \cdot 10^{-7}$ to $2.12 \cdot 10^{-7}$.

By properties of sinusoidal functions, the attention patterns of each head will be well approximated by $0.5 \pm C_j(\cos(w_{k_j}(a + \theta_j)) - \cos(w_{k_j}(b + \theta_j)))$ - the softmax is linear, with an intercept of 0.5, and the weights C_j map each token to a score that is a wave in a single frequency. This exactly gives us the periodic form shown in Figure 9.

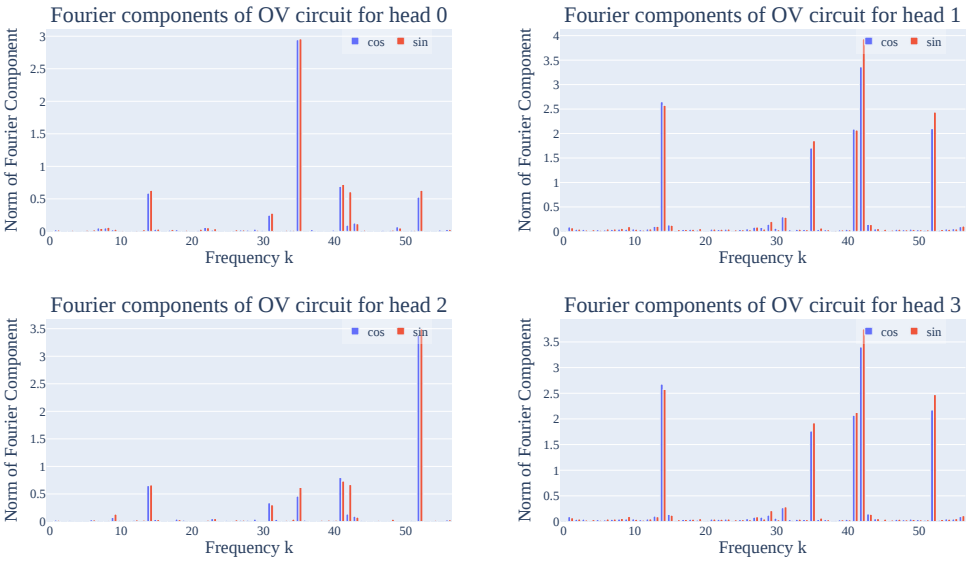

Figure 11: We plot the output of the OV circuit $W_O^j W_V^j x^{(0)}$ in the Fourier basis for each of the four heads $j \in \{0, 1, 2, 3\}$. As with the attention pattern weights C_j in Figure 10, we observe that the only components with significant norm are those corresponding to key frequencies, and that the largest component corresponds to the frequencies of the attention patterns of the attention heads. As attention pattern of heads 1 and 3 are sum to one, but their OV circuits are almost exactly the same and consist of all five key frequencies, this implies that heads 1 and 3 are used to increase the magnitude of key frequencies in the residual stream (Section C.1.3).

Finally, for each head j, we plot the output of the OV circuit $W_O^j W_V^j x^{(0)}$ in the Fourier basis and display the results in Figure 11). The largest component of each head corresponding to the frequency of the attention pattern C_j, with heads 0 and 2 being almost entirely composed of a sines and cosines of a single frequency. On the other hand, the norms for the components of heads 1 and 3 are almost exactly the same, and contain all five key frequencies. As the coefficients of the attention pattern weights have the opposite non-constant components (Table 2, Figure 10), their attention scores sum almost exactly to 1 across all inputs. This implies that heads 1 and 3 are used to output the first order terms $\sin(w_k), \cos(w_k)$ in the five key frequencies. We speculate that this is because of weight decay encouraging the embeddings W_E to be small, causing the network to allocate two of its attention heads to effectively increasing the size of W_E.

Bringing it all together, this implies that attention heads 0 and 2 are approximately computing a degree 2 polynomial of cosines and sines of a single frequency each, while heads 1 and 3 amplify the key frequencies in the residual stream.

C.1.4 Periodicity in the activations of additional neurons

In Figure 12, we display the activations of four more MLP neurons, as a function of the inputs. As with neuron 0, the activations of these neurons are also periodic in the inputs.

C.1.5 Additional grokking figures for mainline run

In Figure 13, we display the accuracy of the model when restricting the model to use only the five key frequencies. As with restricted loss, this *improves* model performance during training.

In Figure 14, we show the coefficients of the five key frequencies in the logits, calculated by regressing the logits against the five $\cos(w_k(a + b - c))$ terms.

In Figure 15, we plot the excluded loss if we exclude each of the five key frequencies (as opposed to all five key frequencies).

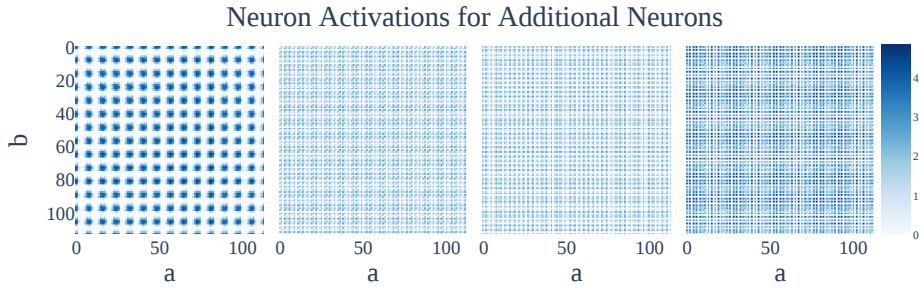

Figure 12: Plots of neuron activations for MLP neurons 1, 2, 3 and 4, for inputs $a, b \in \{0, 1, ..., 112\}$. As with Neuron 0, all of the activation patterns are periodic in both inputs.

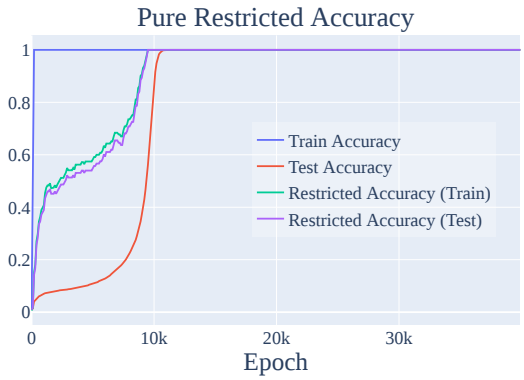

Figure 13: Accuracy when restricting Fourier Components to the five key frequencies. As with restricted loss, this shows that the model figures out how to generalize modulo deleting noise before it removes the noise.

All three of these figures have inflection points corresponding to the relevant phases of grokking, discussed in Section 5.1.

C.2 Additional results from different runs

In this section, we plot relevant figures from other runs, either with the same architecture (Appendix C.2.1) or with different architectures or experimental setups (Appendix C.2.2). Note that in general, while all models learn to use variants of the modular arithmetic algorithm, they use a varying number of *different* key frequencies. In order to find the key frequencies to calculate the excluded and restricted loss, we perform a DFT on the neuron-logit map W_L, then take the frequencies with nontrivial coefficients.[3]

C.2.1 Additional results for different runs with the same architecture

In this section, we provide evidence that all 4 other runs (i.e., random seeds) using the experimental setup of our mainline model also use the Fourier multiplication algorithm, and then confirm that the same phases of grokking also occur on these runs.

[3]One method for getting a general (model-independent) progress measure for this task is to compute the excluded loss for each of the 56 unique frequencies and then take the max. We omit the plots for this variant of the excluded loss as they are broadly similar.